

WARBURTO POINT, Australia

WMO: 946660

Lat: 34.004S

Lon: 137.528E

Elev: 3

StdP: 14.69

Time Zone: 9.50 (AUA)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.5	44.3	25.6	19.6	74.9	28.8	22.8	70.8	27.6	58.8	25.8	59.4	2.5	0	0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.1	101.0	67.4	97.2	66.7	93.2	65.9	72.9	87.4	71.4	85.4	70.0	83.5	8.5	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.7	105.5	78.4	67.1	99.5	77.5	65.6	94.5	76.9	36.5	88.2	35.2	85.0	34.0	83.5	94.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.6	26.6	23.6	DB	39.1	109.1	1.6	3.0	37.9	111.3	36.9	113.0	36.0	114.7	34.8	116.9	(4)
(5)			WB	37.1	78.2	1.9	6.4	35.8	82.8	34.7	86.6	33.6	90.2	32.3	94.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.6	77.5	78.2	74.6	67.8	61.0	55.4	54.2	56.7	61.0	66.1	71.9	75.2
	DBStd	10.38	7.61	7.00	6.92	5.90	4.64	3.94	3.43	4.57	5.90	6.07	7.24	7.20
	HDD50	7	0	0	0	0	0	3	3	1	0	0	0	0
	HDD65	1288	0	0	2	31	143	290	336	262	153	57	12	2
	CDD50	6056	853	789	763	534	342	164	132	210	331	500	656	782
	CDD65	1862	389	369	300	115	20	1	0	6	33	92	218	319
	CDH74	16397	3839	3412	2396	679	69	3	1	31	281	823	1912	2951
	CDH80	7550	1986	1660	1046	221	4	0	0	4	76	304	850	1399
Wind	WSAvg	9.2	12.3	11.6	8.9	6.0	5.4	5.4	6.1	8.0	9.7	11.5	12.3	13.0
Precipitation	PrecAvg	13.8	0.6	0.7	0.6	1.1	1.6	1.8	1.8	1.6	1.6	1.2	0.8	0.7
	PrecMax	21.4	2.7	2.6	2.0	5.4	3.9	4.8	3.6	3.0	4.6	3.8	1.9	2.1
	PrecMin	8.3	0.0	0.0	0.0	0.0	0.2	0.1	0.4	0.3	0.4	0.1	0.0	0.0
	PrecStd	3.0	0.6	0.8	0.6	1.0	0.9	1.0	0.7	0.7	0.9	0.9	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	107.2	105.6	102.1	92.1	80.1	71.5	70.3	78.3	86.7	94.3	101.2	101.9
		MCWB	68.3	72.1	67.6	63.8	59.8	56.3	53.8	57.7	59.1	60.4	65.2	66.1
	2%	DB	100.5	99.5	95.4	86.1	75.6	67.2	66.3	72.4	81.4	87.6	95.0	97.1
		MCWB	67.9	68.7	67.4	61.6	59.4	55.2	52.7	53.9	57.7	59.0	63.8	65.8
	5%	DB	96.0	94.2	90.4	81.3	71.9	64.4	63.4	68.1	76.2	82.7	89.6	92.5
		MCWB	67.1	68.1	65.9	60.9	57.7	54.3	51.9	53.2	55.8	58.0	62.9	65.3
	10%	DB	90.6	89.7	85.7	77.2	68.9	62.0	61.1	64.6	71.9	78.4	84.9	87.7
		MCWB	66.5	67.0	65.6	60.6	56.9	53.1	51.2	52.2	54.9	57.8	62.6	65.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.6	80.9	73.7	68.2	62.8	59.5	58.0	59.9	62.8	64.7	71.1	73.2
		MCDB	89.9	98.1	86.5	79.6	71.7	66.6	62.0	71.3	77.9	76.2	88.3	88.6
	2%	WB	72.4	73.2	71.5	65.7	61.2	57.5	55.0	56.6	60.2	63.0	68.7	71.1
		MCDB	87.1	87.5	84.6	75.5	70.2	64.1	62.1	66.0	74.0	75.2	83.0	84.7
	5%	WB	70.7	71.1	69.5	63.9	60.0	56.0	53.6	54.8	58.1	61.6	66.9	69.5
		MCDB	85.6	85.0	80.5	74.9	68.8	61.5	60.5	63.9	70.8	74.0	80.3	82.3
	10%	WB	69.1	69.6	68.0	62.4	58.7	54.7	52.3	53.4	56.3	60.1	65.2	67.9
		MCDB	84.6	82.9	79.6	73.4	67.0	60.0	59.0	62.1	68.1	72.6	78.2	81.0
Mean Daily Temperature Range	5% DB	MDBR	18.7	18.1	17.1	15.8	14.0	12.9	13.7	14.8	16.8	18.0	18.1	18.0
		MCDBR	26.5	24.9	24.1	22.2	18.5	15.6	17.7	21.0	24.5	27.8	25.4	24.9
		MCWBR	8.2	8.4	6.8	8.0	8.0	7.8	8.6	8.7	8.7	7.5	7.8	8.0
	5% WB	MCDBR	20.6	19.2	17.9	17.3	14.3	12.4	14.7	17.4	20.6	19.1	19.5	18.0
MCWBR		8.0	8.9	7.1	8.4	7.5	7.1	8.0	8.5	8.9	8.3	7.8	8.0	
Clear-Sky Solar Irradiance	taub	0.348	0.346	0.328	0.320	0.310	0.300	0.301	0.308	0.335	0.344	0.348	0.340	
	taud	2.522	2.549	2.589	2.597	2.611	2.624	2.611	2.572	2.484	2.462	2.482	2.531	
	Ebn at Noon	314	308	302	287	273	269	274	287	295	305	312	318	
	Edh at Noon	35	33	30	26	23	21	23	26	32	36	36	35	
All-Sky Solar Radiation	RadAvg	2499	2215	1807	1341	949	787	856	1165	1578	1999	2318	2461	
	RadStd	120	109	76	66	37	39	37	63	107	104	103	97	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-166	+283	+124

Nomenclature: See separate page